

Python Assignment-23

1: Write a Python program using multiprocessing.Pool to calculate the sum of all even numbers from 1 to N for every number from the given list.

Input : Data = [1000000, 2000000, 3000000, 4000000]

Expected Task

For each number N, calculate:

$2 + 4 + 6 + \dots + N$

Expected Output Format

Process ID : 1234

Input Number : 1000000

Sum of Even Numbers : 250000500000

Answer :

```
from multiprocessing import Pool
from multiprocessing import current_process

def SumEven(Values):
    Sum = 0
    for Value in range(2,Values+1,2):
        Sum = Sum + Value
    print("Process ID : ",current_process().pid)
    print("Input Number : ", Values)
    print("Sum of Even Numbers : ",Sum)
    print()
    return Sum

def main():
    print("Enter the list Of numbers : ")
    Values = list(map(int,input().split()))
    print("You Entered List is : ",Values)

    with Pool(4) as p:
        p.map(SumEven,Values)

if __name__ == "__main__":
    main()
```

OUTPUT :

```
GitHub (Workspace)

Code X

PS C:\Users\Sumaney\Desktop\GitHub\Marvellous-Python-Assignments-P229\Assignment-23> python A23Q1.py
Enter the list of numbers :
1 2 3 4 5 6 7 8 9 10
You Entered List is : [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
Process ID : 24960
Input Number : 1
Sum of Even Numbers : 0

Process ID : 24960
Input Number : 2
Sum of Even Numbers : 2

Process ID : 24960
Input Number : 3
Sum of Even Numbers : 2

Process ID : 24960
Input Number : 4
Sum of Even Numbers : 6

Process ID : 24960
Input Number : 5
Sum of Even Numbers : 6

Process ID : 24960
Input Number : 6
Sum of Even Numbers : 12

Process ID : 24960
Input Number : 7
Sum of Even Numbers : 12

Process ID : 24960
Input Number : 8
Sum of Even Numbers : 20

Process ID : 24960
Input Number : 9
Sum of Even Numbers : 20

Process ID : 24960
Input Number : 10
Sum of Even Numbers : 30

PS C:\Users\Sumaney\Desktop\GitHub\Marvellous-Python-Assignments-P229\Assignment-23> 
```

2: Write a Python program using multiprocessing.Pool to calculate the sum of all odd numbers from 1 to N.

Input : Data = [1000000, 2000000, 3000000, 4000000]

Expected Task

For each number N, calculate:

$1 + 3 + 5 + \dots + N$

Expected Output Format

Process ID : 1235

Input Number : 1000000

Sum of Odd Numbers : 250000000000

Answer :

```
from multiprocessing import Pool
from multiprocessing import current_process
```

```
def OddSum(Values):
    Sum = 0
    for Value in range(1,Values+1,2):
        Sum += Value
    print("Process ID : ",current_process().pid)
    print("Input Number : ",Values)
    print("Sum of Odd Numbers : ",Sum)
    print()
    return Sum

def main():
    print("Enter the list of Numbers : ")
    Values = list(map(int,input().split()))
    print("You Entered List of Numbers is : ",Values)

    with Pool(4) as P:
        P.map(OddSum,Values)

if __name__ == "__main__":
    main()
```

OUTPUT :

```
GitHub (Workspace)

Code X

PS C:\Users\Sumaney\Desktop\GitHub\Marvellous-Python-Assignments-P229\Assignment-23> python A23Q2.py
Enter the list of Numbers :
1 2 3 4 5 6 7 8 9 10
You Entered List of Numbers is : [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
Process ID : 35188
Input Number : 1
Sum of Odd Numbers : 1

Process ID : 35188
Input Number : 2
Sum of Odd Numbers : 1

Process ID : 35188
Input Number : 3
Sum of Odd Numbers : 4

Process ID : 35188
Input Number : 4
Sum of Odd Numbers : 4

Process ID : 35188
Input Number : 5
Sum of Odd Numbers : 9

Process ID : 35188
Input Number : 6
Sum of Odd Numbers : 9

Process ID : 35188
Input Number : 7
Sum of Odd Numbers : 16

Process ID : 35188
Input Number : 8
Sum of Odd Numbers : 16

Process ID : 35188
Input Number : 9
Sum of Odd Numbers : 25

Process ID : 35188
Input Number : 10
Sum of Odd Numbers : 25

PS C:\Users\Sumaney\Desktop\GitHub\Marvellous-Python-Assignments-P229\Assignment-23> 
```

3: Write a program that counts how many even numbers exist between 1 and N using Pool.map().

Input : Data = [1000000, 2000000, 3000000, 4000000]

Expected Output Format

Process ID : 1236

Input Number : 1000000

Even Number Count : 500000

Answer :

```
from multiprocessing import Pool
from multiprocessing import current_process

def EvenCount(Values):
    Count = 0
    for Value in range(2,Values+1,2):
        Count = Count + 1
    print("Process Id : ",current_process().pid)
    print("Input Number : ",Values)
```

```

    print("Even Number Count : ",Count)
    print()
    return Count

def main():
    print("Enter the list of Numbers")
    Values = list(map(int,input().split()))
    print("You Entered number of list is : ",Values)

    with Pool(4) as P:
        Result = P.map(EvenCount,Values)

if __name__ == "__main__":
    main()

```

OUTPUT :


← → GitHub (Workspace)

Code

X

```

PS C:\Users\Sumaney\Desktop\GitHub\Marvellous-Python-Assignments-P229\Assignment-23> python A23Q3.py
Enter the list of Numbers
1 2 3 4 5 6 7 8 9 10
You Entered number of list is : [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
Process Id : 7448
Input Number : 1
Even Number Count : 0

Process Id : 7448
Input Number : 2
Even Number Count : 1

Process Id : 7448
Input Number : 3
Even Number Count : 1

Process Id : 7448
Input Number : 4
Even Number Count : 2

Process Id : 7448
Input Number : 5
Even Number Count : 2

Process Id : 7448
Input Number : 6
Even Number Count : 3

Process Id : 7448
Input Number : 7
Even Number Count : 3

Process Id : 7448
Input Number : 8
Even Number Count : 4

Process Id : 7448
Input Number : 9
Even Number Count : 4

Process Id : 7448
Input Number : 10
Even Number Count : 5

PS C:\Users\Sumaney\Desktop\GitHub\Marvellous-Python-Assignments-P229\Assignment-23>

```

4: Write a program that counts how many odd numbers exist between 1 and N.

Input : Data = [1000000, 2000000, 3000000, 4000000]

Expected Output Format

Process ID : 1237

Input Number : 1000000

Odd Number Count : 500000

Answer :

```
from multiprocessing import Pool
from multiprocessing import current_process

def OddCount(Values):
    Count = 0
    for Value in range(1,Values+1,2):
        Count = Count + 1
    print("Process Id : ",current_process().pid)
    print("Input Number : ",Values)
    print("Odd Number Count : ",Count)
    print()
    return Count

def main():
    print("Enter the list of Numbers")
    Values = list(map(int,input().split()))
    print("You Entered number of list is : ",Values)

    with Pool(4) as P:
        Result = P.map(OddCount,Values)

if __name__ == "__main__":
    main()
```

OUTPUT :

```
GitHub (Workspace)

Code X

PS C:\Users\Sumaney\Desktop\GitHub\Marvellous-Python-Assignments-P229\Assignment-23> python A23Q4.py
Enter the list of Numbers
1 2 3 4 5 6 7 8 9 10
You Entered number of list is : [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
Process Id : 21824
Input Number : 1
Odd Number Count : 1

Process Id : 21824
Input Number : 2
Odd Number Count : 1

Process Id : 21824
Input Number : 3
Odd Number Count : 2

Process Id : 21824
Input Number : 4
Odd Number Count : 2

Process Id : 21824
Input Number : 5
Odd Number Count : 3

Process Id : 21824
Input Number : 6
Odd Number Count : 3

Process Id : 21824
Input Number : 7
Odd Number Count : 4

Process Id : 21824
Input Number : 8
Odd Number Count : 4

Process Id : 21824
Input Number : 9
Odd Number Count : 5

Process Id : 21824
Input Number : 10
Odd Number Count : 5

PS C:\Users\Sumaney\Desktop\GitHub\Marvellous-Python-Assignments-P229\Assignment-23>
```

5: Write a program that calculates factorials of multiple numbers simultaneously using multiprocessing.Pool.

Input

Data = [10, 15, 20, 25]

Expected Task

For every N, calculate:

N!

Expected Output Format

Process ID : 1240

Input Number : 20

Factorial : 243290200817664000

Answer :

```
#
=====
===
```

```

# Author   : codingwithamar@gmail.com
# File     : A23Q5.py
# Path     : Marvellous-Python-Assignments-P229\Assignment-23\A23Q5.py
# Subject/Question : 5: Write a program that calculates factorials of
multiple
# numbers simultaneously using multiprocessing.Pool.
# Description : Input Data = [10, 15, 20, 25]
# Expected Task : For every N, calculate: N!
# Expected Output Format : Process ID : 1240
#                               Input Number : 20
#                               Factorial : 2432902008176640000
#
=====
===

```

```

from multiprocessing import Pool
from multiprocessing import current_process

def Factorial(iValues):
    Sum = 1
    for i in range(1,iValues+1):
        Sum = Sum * i
    print("Process ID : ",current_process().pid)
    print("Input Number : ",iValues)
    print("Factorial : ",Sum)
    print()
    return Sum

def main():
    print("Enter the list of Numbers")
    Values = list(map(int,input().split()))
    print("Your Entered List is : ",Values)

    with Pool(4) as P:
        P.map(Factorial,Values)

if __name__ == "__main__":
    main()

```

OUTPUT :



GitHub (Workspace)

Code X

```
PS C:\Users\Sumaney\Desktop\GitHub\Marvellous-Python-Assignments-P229\Assignment-23> python A23Q5.py
Enter the list of Numbers
1 2 3 4 5 6 7 8 9 10
Your Entered List is : [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
Process ID : 9352
Input Number : 1
Factorial : 1

Process ID : 9352
Input Number : 2
Factorial : 2

Process ID : 9352
Input Number : 3
Factorial : 6

Process ID : 9352
Input Number : 4
Factorial : 24

Process ID : 9352
Input Number : 5
Factorial : 120

Process ID : 9352
Input Number : 6
Factorial : 720

Process ID : 9352
Input Number : 7
Factorial : 5040

Process ID : 9352
Input Number : 8
Factorial : 40320

Process ID : 9352
Input Number : 9
Factorial : 362880

Process ID : 9352
Input Number : 10
Factorial : 3628800

PS C:\Users\Sumaney\Desktop\GitHub\Marvellous-Python-Assignments-P229\Assignment-23> |
```